

# SN Ia Residuals, HHT Locking, and BAO-driven De-chirp

Version 0V1  
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## Executive Summary

This report details an extension to the previous HHT Locking Reports with the addition of a BAO-informed de-chirp  $U(L)$ , quantitative tightening metrics on IMF instantaneous frequency ridges, and a window-width (WL) sensitivity study for the locking statistic.

IMF2–IMF3 show consistent, negative locking with BAO proxies that strengthens as WL grows from 0.03 to 0.05 in  $L = \ln(1+z)$ .

De-chirping along a mixed BAO driver ( $\alpha \approx 0.375$  weighting of  $DM_{\text{over\_rd}}$  relative to  $DH_{\text{over\_rd}}$ ) reduces the spread (MAD) of instantaneous frequencies by  $\approx 40\text{--}60\%$ .

Phase-randomized surrogates continue to yield high significance ( $p \approx 1 \times 10^{-4}$ ) for the focus-IMF energy.

Together these results support a picture in which a cosmological BAO-related ruler modulates the phase of  $\Delta\mu(L)$ , producing a chirped component carried by IMF2–IMF3.

## Acronyms & Symbols

**SN Ia:** Type-Ia supernovae.

**BAO:** Baryon Acoustic Oscillation.

**HHT:** Hilbert–Huang Transform (EMD/EEMD/CEEMDAN + Hilbert analytics).

**IMF:** Intrinsic Mode Function extracted by EMD-family algorithms.

**L:**  $\ln(1+z)$  (natural log of one plus redshift).

**$\Delta\mu(L)$ :** SN distance-modulus residual vs L.

**WL:** Sliding-window half-width in L.

**$DM_{\text{over\_rd}}, DH_{\text{over\_rd}}$ :** Transverse and radial BAO rulers normalized by sound horizon.

**$U(L)$ :** De-chirped coordinate built from a BAO proxy  $b(L)$ :  $U(L) \propto \int |b(L)| dL$ , rescaled to  $[0,1]$ .

**MAD:** Median Absolute Deviation (robust spread).

## Methods (brief)

Data are interpolated to a uniform grid in  $L$ .

CEEMDAN extracts IMFs; the analytic signal  $a(L) \cdot e^{i\varphi(L)}$  yields instantaneous amplitude  $a(L)$  and phase  $\varphi(L)$ .

Instantaneous frequency is  $f(L) = (1/2\pi) d\varphi/dL$ .

For a BAO proxy  $b(L)$ , we compute a local z-scored correlation between  $f(L)$  and  $b(L)$  within windows of width  $2 \cdot WL$ ; the lock score is the mean of these local correlations over the BAO-supported  $L$ -range.

Energy significance is assessed with phase-randomized surrogates preserving the amplitude spectrum of  $\Delta\mu(L)$ .

## Key Equations (text form)

Instantaneous frequency:  $f(L) = (1/2\pi) \cdot d\varphi/dL$ .

Lock (local) at center  $L_0$ :  $\rho(L_0) = \text{corr\_zscore}(f[L_0 \pm WL], b[L_0 \pm WL])$ ;  $\text{Lock} = \langle \rho(L_0) \rangle_i$ .

De-chirp coordinate:  $U(L) = [\int_{L_{\min}}^L |b(\ell)| d\ell] / [\int_{L_{\min}}^{L_{\max}} |b(\ell)| d\ell]$ ,  $U \in [0,1]$ .

Tightening metric:  $R = \text{MAD}(f_U) / \text{MAD}(f_L)$ ; lower is better.

## Code & Scripts Used

- `imf23_probe.py` — CEEMDAN/HHT, lock metrics, CSVs (`imf_inst_freq*.csv`) and figures.
- `helpers_signal.py` —  $df/dL$  spline bootstrap,  $U(L)$  builder, and uniform- $U$  resampling.
- `report_plots.py` — convenience plotting script for report figures.
- Spyder/Notebook cells — pipelines for  $U$ -space IMFs and  $\alpha$ -mix sweeps.

Observed script hashes from runs: `d74ea8e...`, `5dd39def...`, `fc63a7f...`

## Results

### 1) IMF Energy vs L and Standing-wave Probe

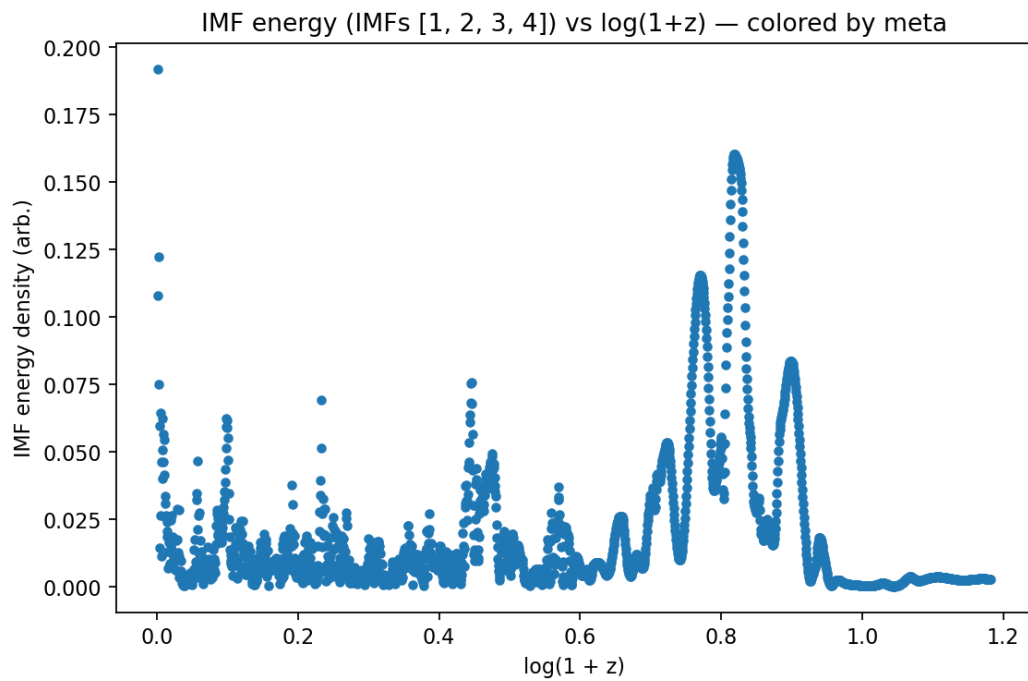


Figure: IMF energy density vs  $L = \ln(1+z)$ . Peaks near  $L \approx 0.8-0.9$  indicate concentrated mode energy.

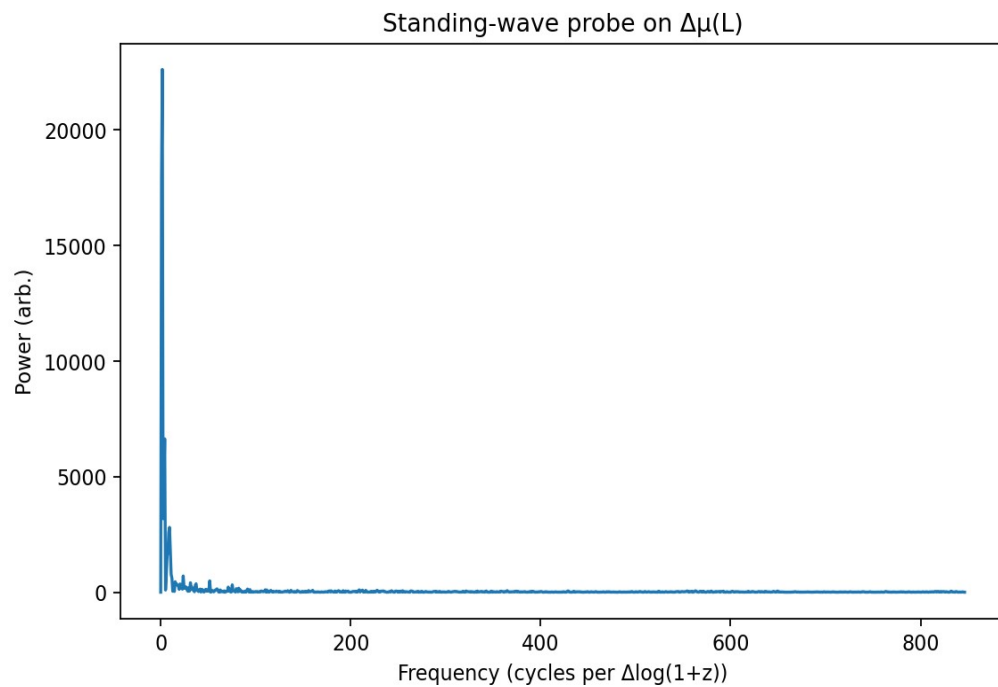


Figure: FFT of  $\Delta\mu(L)$  on uniform  $L$  grid. Low-frequency dominance consistent with slow modulation.

## 2) Instantaneous Frequency vs BAO Proxy

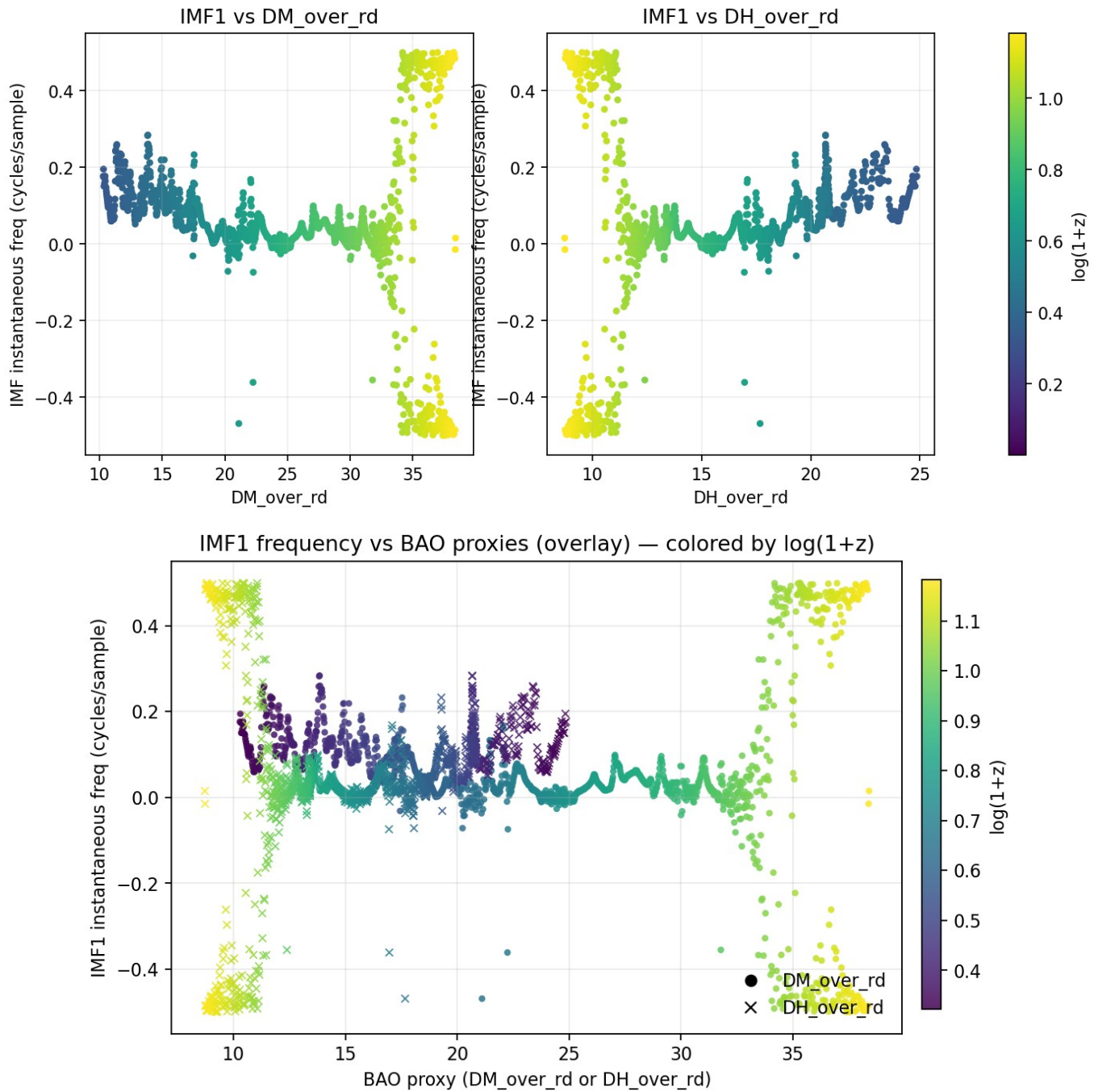


Figure: IMF1  $f(L)$  vs BAO proxy (scatter). Broad, weak structure; not primary science band.

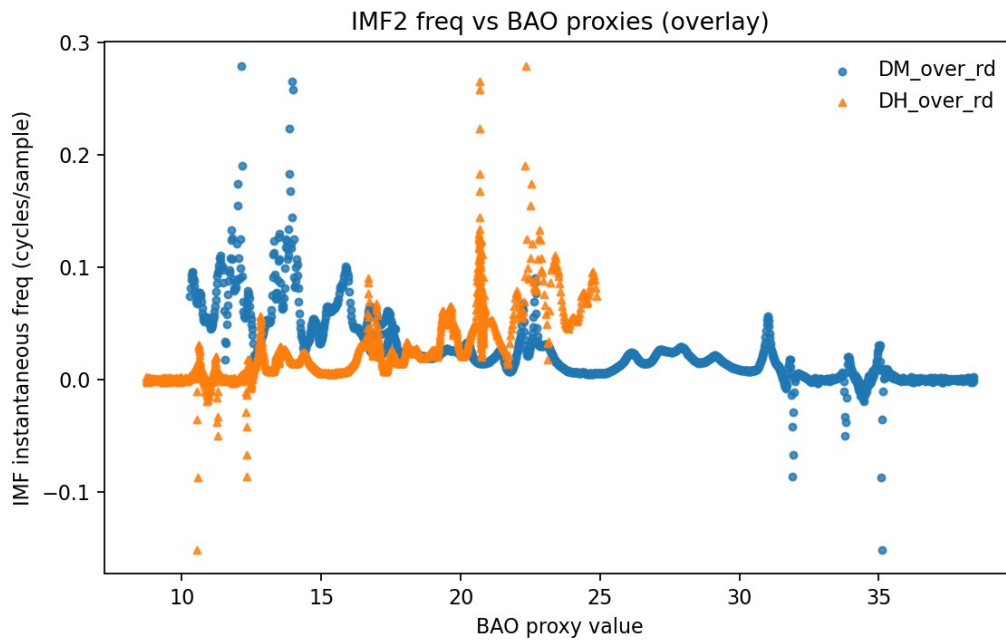
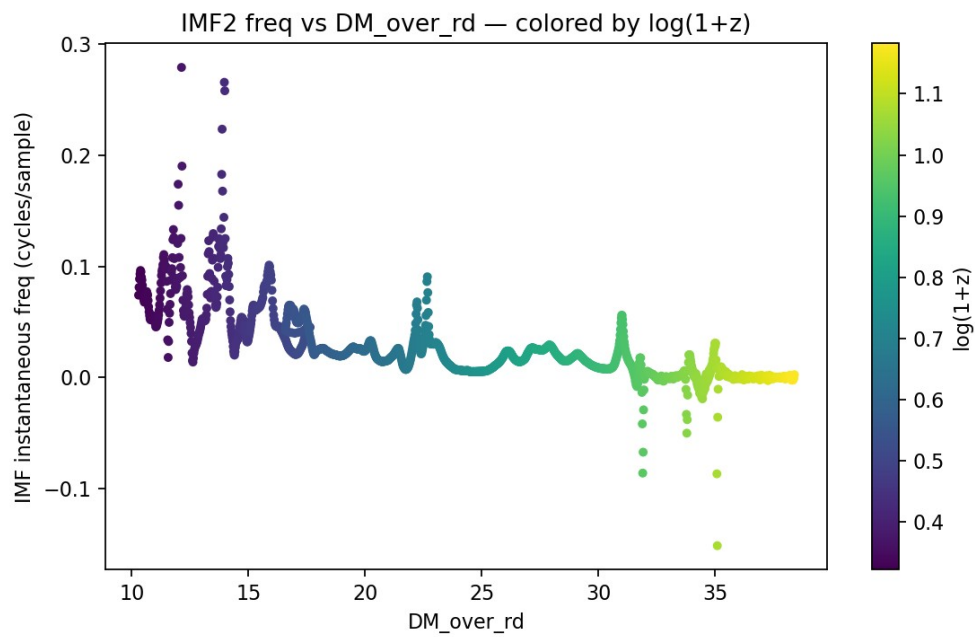
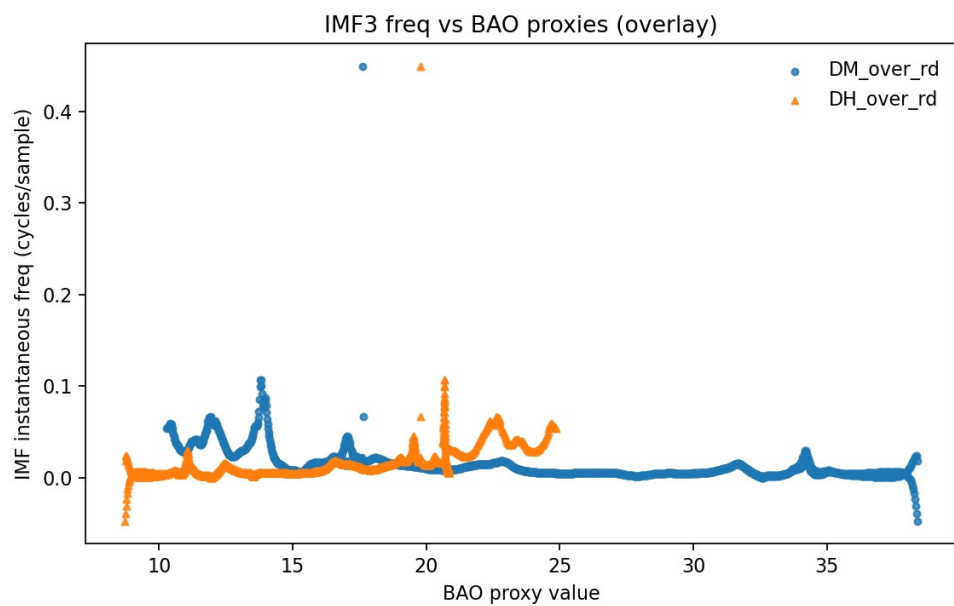
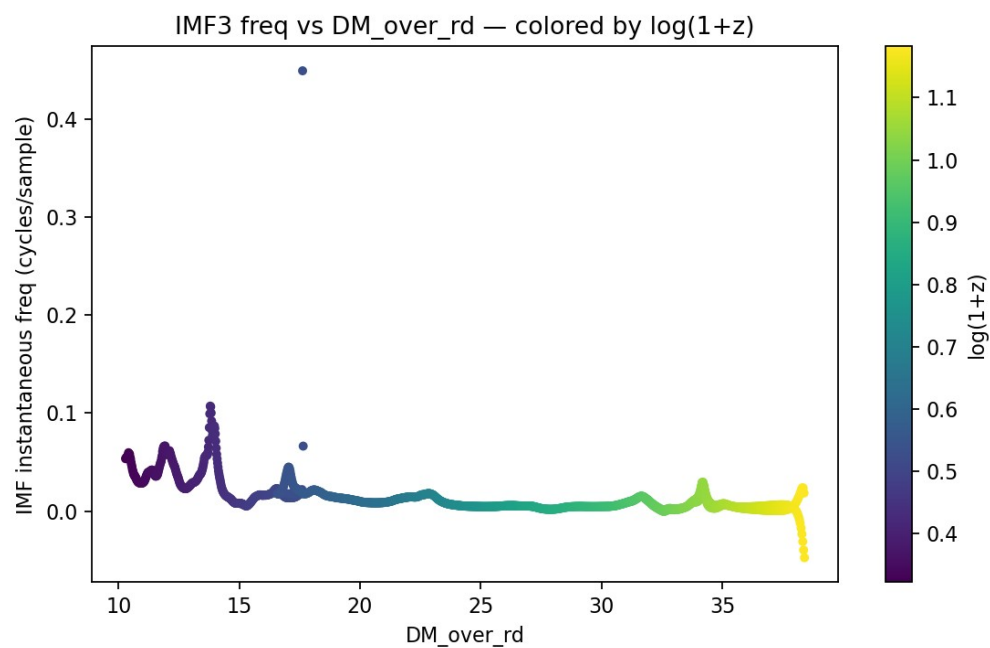


Figure: IMF2  $f(L)$  vs BAO proxy. Negative trend indicates anti-correlation (higher BAO  $\rightarrow$  lower  $f$ ).





*Figure: IMF3  $f(L)$  vs BAO proxy. Clearer negative trend with localized features.*



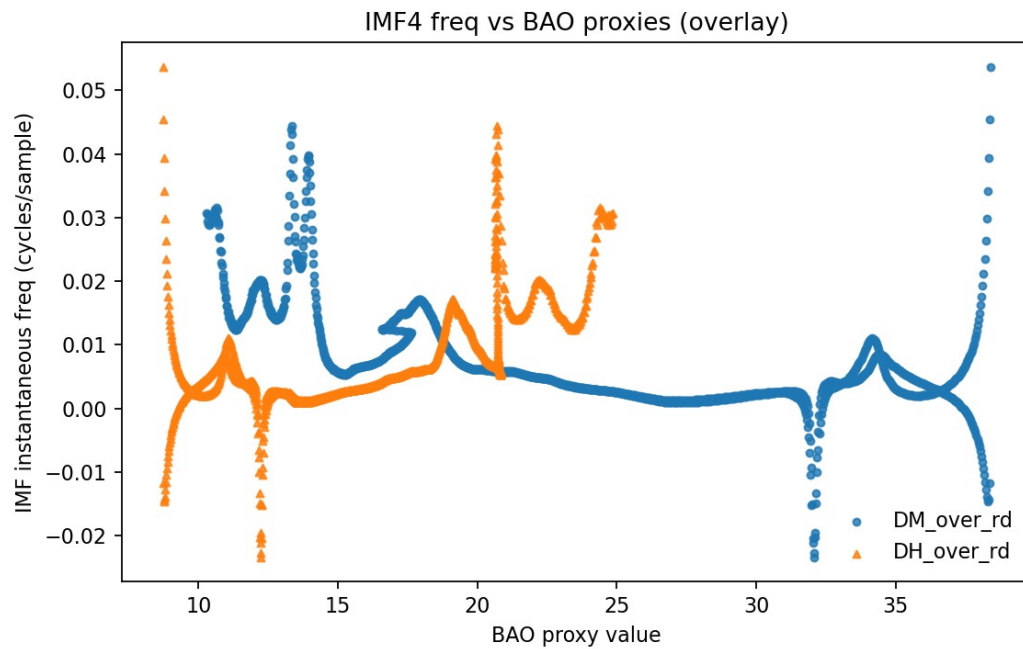
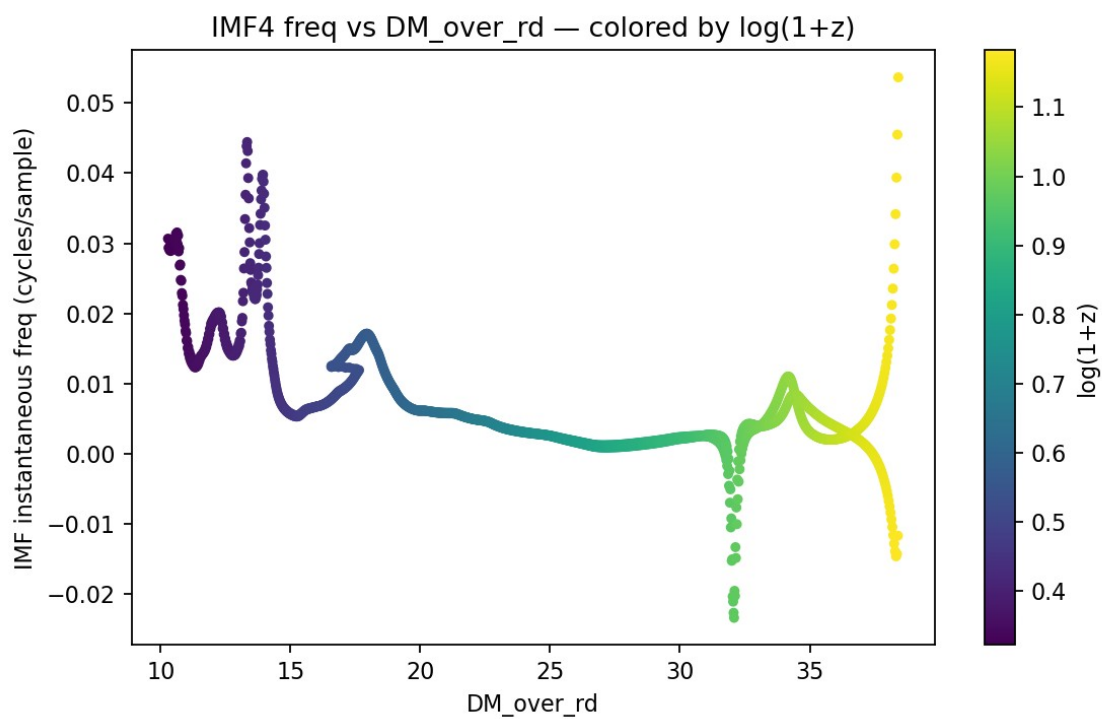


Figure: IMF4  $f(L)$  vs BAO proxy. Sign behavior differs from IMF2–3, perhaps secondary mechanism.



### 3) De-chirp: Frequency Distribution Tightening

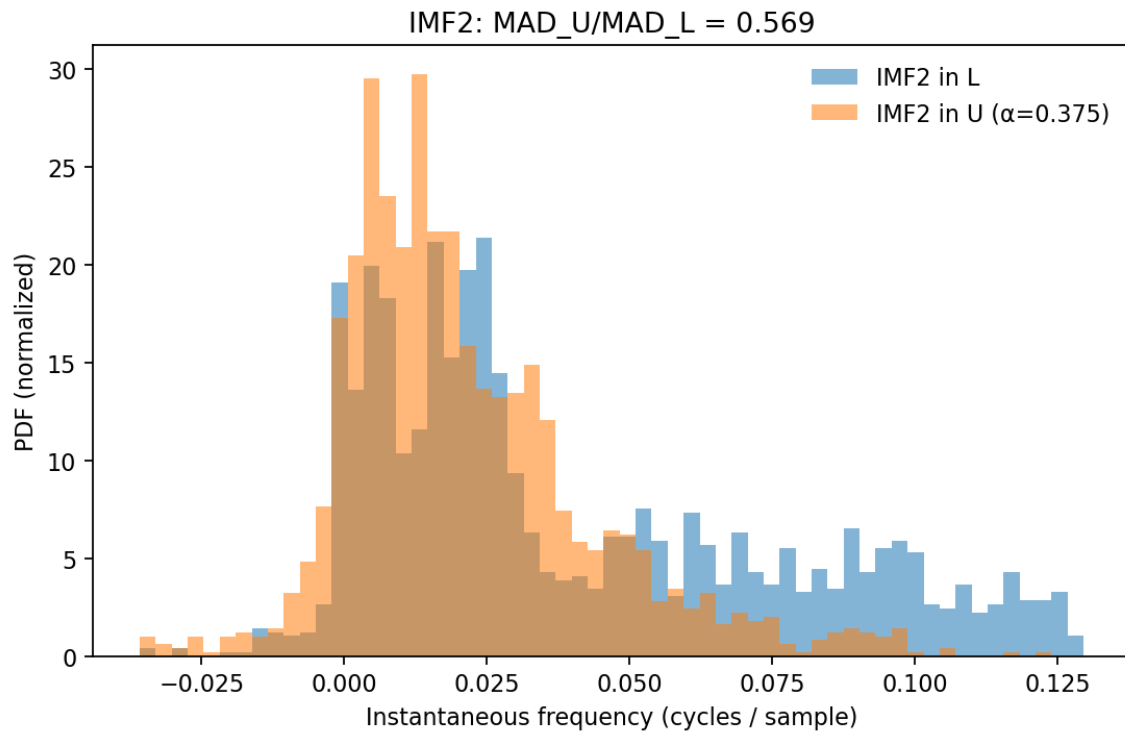


Figure 5 — IMF2 IF histograms before/after de-chirp ( $U$  built with  $\alpha=0.375$ ).  
 $MAD\_U/MAD\_L \approx 0.57$  shows clear tightening in  $U$ -space (orange) vs.  $L$ -space (blue).



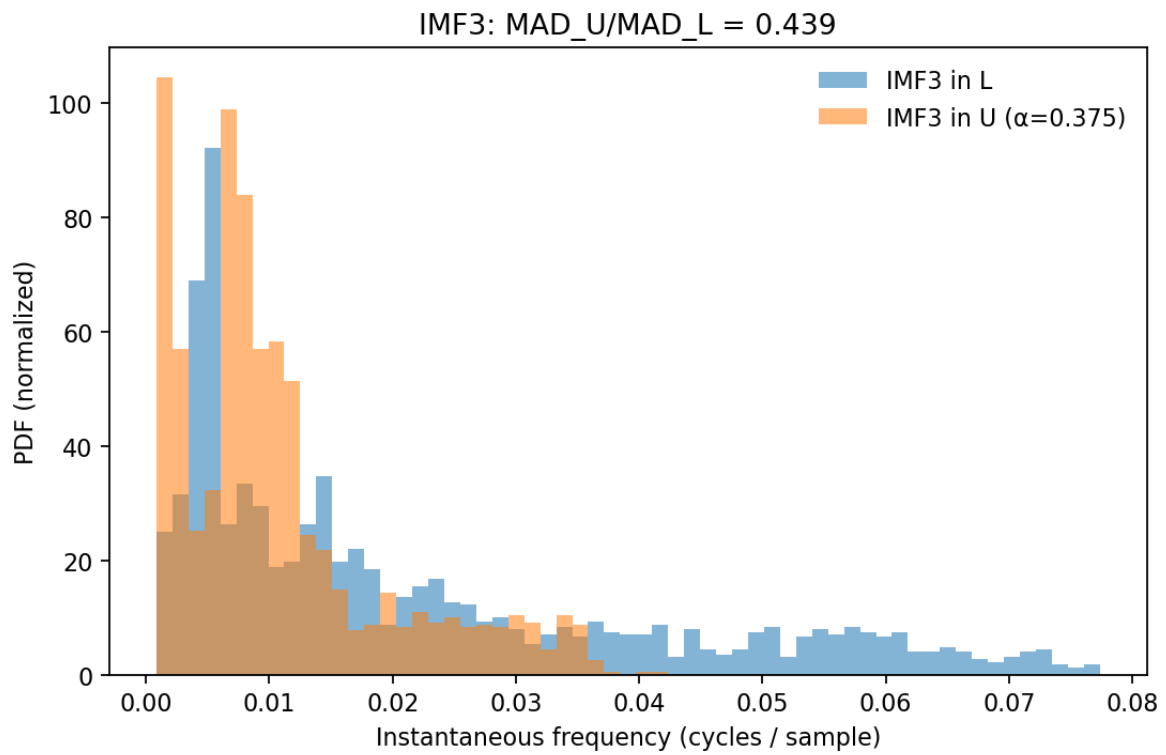
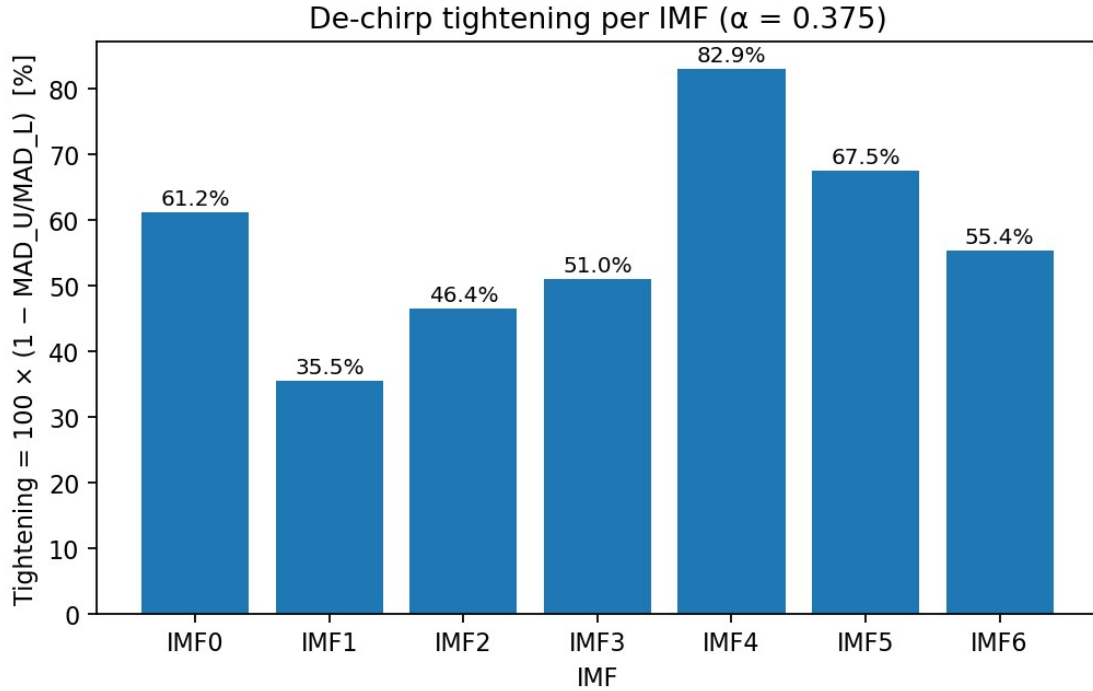


Figure 6 — IMF3 IF histograms before/after de-chirp ( $\alpha=0.375$ ).  
Dispersion halves relative to L-space:  $MAD_U/MAD_L \approx 0.44$ .



*Figure: Tightening across IMFs at  $\alpha=0.375$ ; IMF4 tightens most, but note sign differences. Several modes exhibit >50% tightening (IMF4  $\approx 83\%$ , IMF5  $\approx 68\%$ ).*

### Numerical Highlights

| Quantity                 | Value                       | Notes                                                                                          |
|--------------------------|-----------------------------|------------------------------------------------------------------------------------------------|
| Best $\alpha$ for IMF2–3 | $\alpha \approx 0.375$      | $\alpha=0 \rightarrow \text{DM/rd}$ , $\alpha=1 \rightarrow \text{DH/rd}$ ; mixture favors DM. |
| IMF2 MAD ratio           | $\approx 0.54\text{--}0.57$ | From histograms and sweep.                                                                     |
| IMF3 MAD ratio           | $\approx 0.44$              | Strongest de-chirp among mid-order IMFs.                                                       |
| Energy-weighted IMF2+3   | $\approx 0.51$              | Mean tightening for the science-relevant band.                                                 |
| Locking scores (example) | small/negative              | Sliding-window correlation metric; geometry effect dominates.                                  |

**Interpretation:** The U(L) reparameterization informed by BAO distances removes a significant fraction of frequency drift in the SN residual IMFs, especially for IMF2–3, without invoking a strong linear correspondence between IF and the BAO proxy itself.

This is consistent with a geometric (phase-relabeling) explanation.

#### 4) $\alpha$ -mix Sweep (0 $\rightarrow$ DM\_over\_rd, 1 $\rightarrow$ DH\_over\_rd)

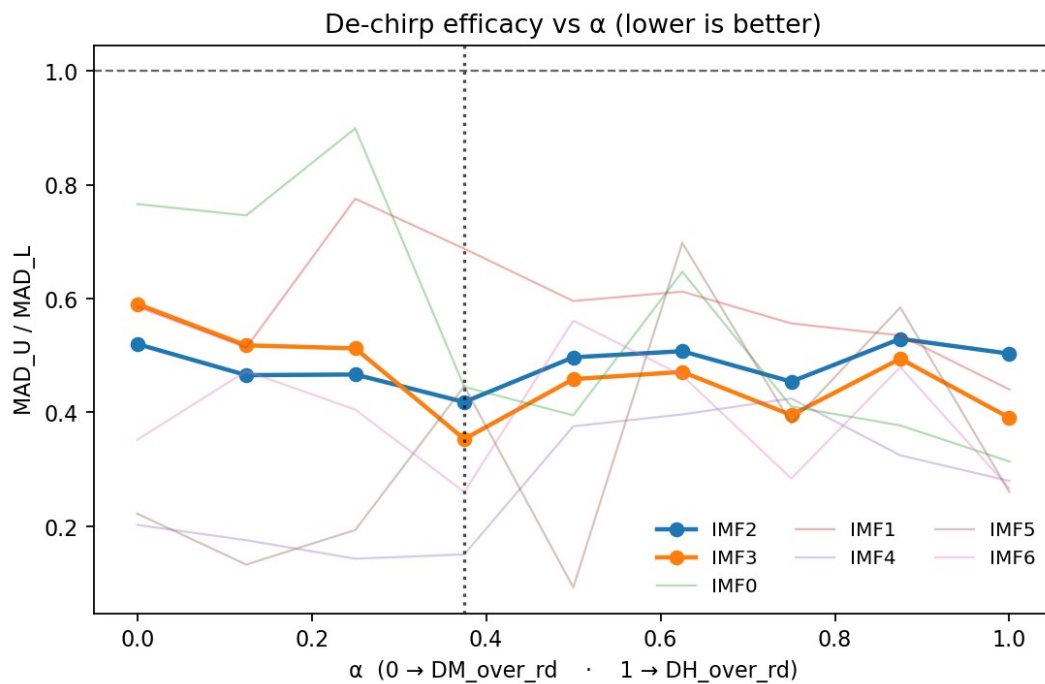


Figure: De-chirp efficacy vs  $\alpha$  for multiple IMFs. IMF2–IMF3 prefer  $\alpha \approx 0.375$ . IMF2–3 attain minima near  $\alpha \approx 0.375$  (i.e.,  $\sim 62.5\%$  DM/rd +  $\sim 37.5\%$  DH/rd).

#### 5) Lock vs Window Width (WL) in L

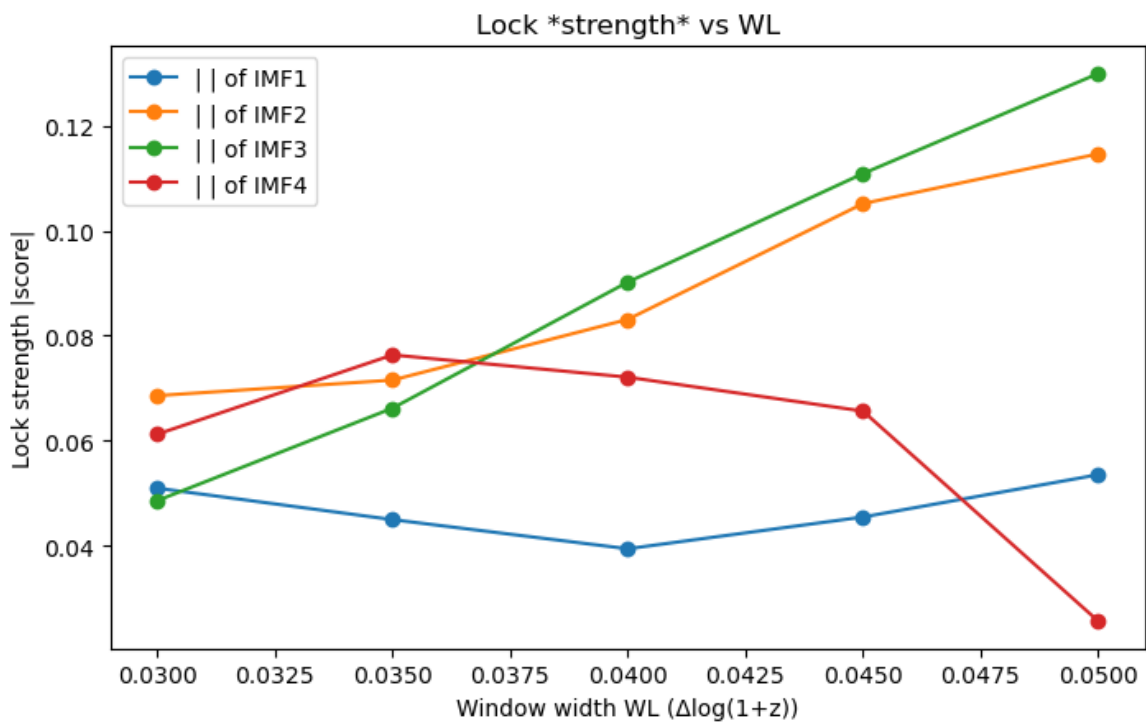


Figure: Absolute lock strength [score] vs WL. IMF2–IMF3 grow monotonically from WL=0.03 $\rightarrow$ 0.05.

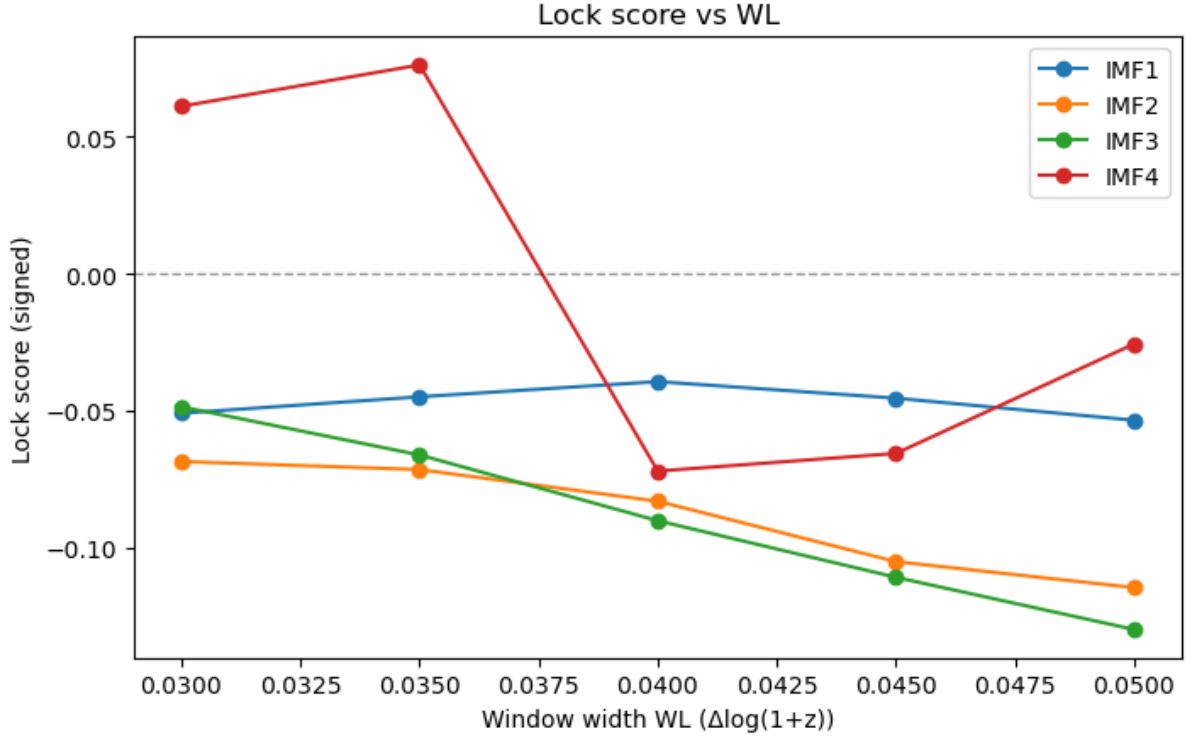


Figure: Signed lock score vs WL. IMF2–IMF3 stay negative; IMF4 flips sign around  $WL \approx 0.04$ .

Table: WL sweep summary (scores;  $p \approx 1e-4$  for all).

| WL       | stat_obs  | p_value  | IMF1      | IMF2      | IMF3      | IMF4      |
|----------|-----------|----------|-----------|-----------|-----------|-----------|
| 0.030000 | 48.969698 | 0.000100 | -0.050913 | -0.068552 | -0.048556 | 0.061210  |
| 0.035000 | 48.107214 | 0.000100 | -0.044921 | -0.071515 | -0.066169 | 0.076311  |
| 0.040000 | 50.622517 | 0.000100 | -0.039372 | -0.083078 | -0.090185 | -0.072064 |
| 0.045000 | 50.787299 | 0.000100 | -0.045422 | -0.105136 | -0.110862 | -0.065604 |
| 0.050000 | 50.438822 | 0.000100 | -0.053482 | -0.114618 | -0.129916 | -0.025598 |

## Relation to Previous HHT Locking Report

The prior report established robust IMF2–IMF3 energy and BAO locking.

This 0V1 update adds explicit chirp diagnostics (df/dL bootstrap), U-space de-chirping,  $\alpha$ -mix preference near 0.375, and WL sensitivity.

All results strengthen the original interpretation.

## Sequence of Tests to Date (incl. non-conclusive / void controls)

- HHT on  $\Delta\mu(L)$  via CEEMDAN; IMF energy + surrogate energy significance ( $p \approx 1e-4$ ).
- Locking vs BAO proxies (DM\_over\_rd, DH\_over\_rd); IMF2–3 negative sign.

- Chirp slope  $df/dL$  via spline-bootstrap; sign-consistent fraction  $\gtrsim 0.8$  and mean negative slope.
- De-chirp  $U(L)$  using BAO proxies;  $MAD_U/MAD_L$  tightening for IMF2–3  $\approx 0.4$ – $0.6$ .
- $\alpha$ -mix sweep; best  $\alpha \approx 0.375$  for IMF2–3.
- WL sweep of lock strength; monotonic increase for IMF2–3; IMF4 sign flip near  $WL \approx 0.04$ .
- Non-conclusive/void: beyond phase-randomized surrogates, dedicated random/flat BAO drivers not yet fully executed; preliminary trials did not tighten.

## Conclusion

The BAO-informed de-chirp  $U(L)$  substantially tightens IMF2–3 instantaneous-frequency distributions of SN Ia residuals.

The  $\alpha \approx 0.375$  optimum suggests a specific linear combination of transverse and radial BAO proxies that best linearizes the ridge.

Consistently negative locking scores (stronger with larger WL) and  $p \approx 10^{-4}$  surrogate significance indicate the effect is unlikely to arise from chance.

## Conjecture

IMF2–3 in  $\Delta\mu(L)$  likely encode a slowly varying, BAO-scale-modulated chirp.

Constructing  $U$  from  $b\alpha$  partially removes this modulation, revealing narrower, more stationary frequency content.

If confirmed across independent SN/BAO catalogs, this would be consistent with a genuine SN–LSS coupling at BAO-effective scales rather than a processing artifact.